PharmAchieve

SYSTEMIC LUPUS ERYTHEMATOSUS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION…

- Providing Care: Clinical Care
- Providing Care: Distribution
- Knowledge & Expertise
- Communication & Collaboration
- Leadership & Stewardship
- Professionalism

LEGEND

ACEi	Angiotensin-Converting Enzyme inhibitor
ACR	American College Of Rheumatology
AE	Adverse Effect
ANA	Antinuclear Antibodies
AZA	Azathioprine
BID	Bis In Die (Twice Daily)
BLyS	B-lymphocyte Stimulator
CBC	Complete Blood Count
CNI	Calcineurin Inhibitors
CS	Corticosteroids
CVD	Cardiovascular Disease
CYC	Cyclophosphamide
dsDNA	Double-stranded DNA
GI	Gastrointestinal
HCQ	Hydroxychloroquine
HTN	Hypertension
IM	Intramuscular
IV	Intravenous
LFT	Liver Function Test
MI	Myocardial Infarction
MMR	Measles, Mumps, Rubella
MTX	Methotrexate
NSAIDs	Non-steroidal Anti-inflammatory Drugs
PML	Progressive Multifocal Leukoencephalopathy
PO	Per Os (By Mouth)
Q	Quaque (Every)
SLE	Systemic Lupus Erythematosus
SPF	Sun Protection Factor
TID	Ter In Die (Three Times A Day)
UTI	Urinary Tract Infection
VTE	Venous Thromboembolism

DEFINITIONS

WHAT IS SYSTEMIC LUPUS ERYTHEMATOSUS?

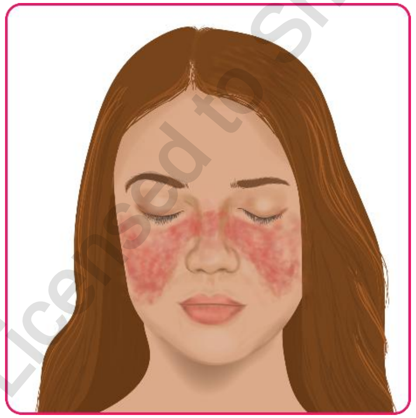
- **SLE** is an autoimmune disease of unknown cause in which the body's immune system attacks healthy body tissue
- It is chronic in nature and causes pain and inflammation in any part of the body

When the immune system is functioning normally, it makes proteins called antibodies that protect against pathogens (e.g. viruses). SLE is characterized by the presence of antibodies against a person's own proteins (autoantibodies). The most common autoantibody found is anti-nuclear antibodies.

4 Types of lupus: SLE, drug-induced lupus, neonatal lupus, cutaneous lupus

CLINICAL PRESENTATION

- Joint pain and swelling (the joints of the fingers, wrists, hands and knees are affected)
- Swollen lymph nodes
- Fever (with no other cause)
- Cardiac involvement (e.g. chest pain) and vascular manifestations
- Hair loss
- Mouth sores/ulcers
- Skin rash ("butterfly" rash seen over the cheeks and bridge of nose; can be widespread; may worsen in sunlight)



- General discomfort/ uneasiness/ malaise
- Prolonged or extreme fatigue (80-100%)
- Arthritis
- Sensitivity to sunlight
- Thromboembolic disease
- Renal involvement (e.g. lupus nephritis)
- Weight changes
    - Loss= change of appetite, medication AE (e.g. HCQ), GI disease
    - Gain= medication AE (e.g. CS), kidney problems

Symptom type and duration vary between patients

RISK FACTORS

Family history

Gender

- More common in women than men

Age

- 15-44 years old

African Americans, Hispanics, Native Americans and Asians

DRUG-INDUCED LUPUS

- Drug-induced lupus is brought on by an overreaction to a medication
- Medications definitively associated with lupus include:
  - Chlorpromazine
  - Hydralazine
  - Isoniazid
  - Methyldopa
  - Minocycline
  - Procainamide
  - Quinidine

NOTE: antitumor necrosis factor-alpha therapy and interferon-alfa have also been implicated in drug-induced lupus. Sub-acute cutaneous lupus has been associated with calcium channel blockers, ACE-inhibitors, hydrochlorothiazide, leflunomide, and terbinafine

GOALS OF THERAPY

- ✓ Relieve the symptoms of SLE and prevent damage and complications from SLE
- ✓ Control autoimmune activation and inflammation
- ✓ Improve quality of life and long-term survival
- ✓ In patients with severe disease: induce remission, followed by maintenance

NON-PHARMACOLOGICAL MEASURES

Patient education

Maintain immunizations

- Maintain immunizations, advise on receiving annual influenza vaccination, avoid live vaccines

Avoid prolonged sun exposure

- Wear protective hat and clothing, and use sunscreen with SPF ≥30

Eat a heart-healthy diet and exercise regularly

Smoking Cessation

NOTE

Due to reduced sun exposure, patients are at increased risk for osteoporosis. Patients should supplement with vitamin D (at least 1000 units daily) and calcium (if their intake of elemental Ca++ is not at least 1200 mg/day total from all sources)

PHARMACOLOGICAL MEASURES

Topical Therapies

NSAIDs

Anti-malarials

Corticosteroid-Sparing Agents

Corticosteroids

TOPICAL THERAPIES

- Mild local rashes may be sufficiently treated with topical corticosteroids or calcineurin inhibitors (e.g., tacrolimus, pimecrolimus)
- Patients with more refractory skin diseases may require systemic therapy

NON-STEROIDAL ANTI-INFLAMMATORY AGENTS (NSAIDS)

- Indication: As needed for joint pain (in patients with arthritis) and pleuritic chest pain (in patients with pleuritis or pericarditis)
  - All NSAIDs increase risk of MI or stroke
  - In some patients, high doses of NSAIDs may be associated with the development of aseptic meningitis
  - Should be limited to short periods of time for symptom control

ANTI-MALARIALS

First-Line

- Hydroxychloroquine (HCQ): 1st line unless contraindicated
  - Improves photosensitive rashes, arthritis and fatigue
  - Can be combined with corticosteroids and immunosuppressants

Advantages	Disadvantage
• Can reduce accumulated damage if used early enough • Lipid and glucose-lowering effects • Reductions in blood clots	• Withdrawal of these agents has been linked to disease flares in stable patients • Need to have regular ophthalmologic assessments to prevent irreversible retinopathy • Risk factors include: high-dose, therapy >5-7 years, existing liver/kidney disease, advanced age, obesity, pre-existing ophthalmologic disease

GLUCOCORTICOIDS (INTRAARTICULAR, IM, IV, PO)

- Used as ‘bridging therapy’ to obtain rapid control of acute inflammation (induce remission) and for flare ups as needed
- Combined with antimalarials
- Dosing should be based on the type and severity of organ involvement
- Myalgias, arthralgias/arthritis, fatigue, low-grade fever: low-dose corticosteroids
- Pleuritis/pericarditis: moderate to high doses of prednisone (0.5-1 mg/kg/day or more)
- Life/organ-saving: consider IV “pulse” methylprednisolone (250 – 1000 mg daily for 1 to 3 days) then move down to PO 0.3 – 0.5 mg/kg/day
    - In some cases, higher initial doses may be given (0.7-0.8 mg/kg/day)
- In all circumstances: maintenance doses should be kept at ≤ 5 mg/day (prednisone equivalent), or completely withdrawn, when possible

OTHER PHARMACOLOGICAL THERAPIES

Second-Line

Choose agent based on patient-specific factors (e.g. symptoms, medications, comorbidities). These agents are also steroid-sparing and should be initiated early to minimize glucocorticoid-related toxicity

Azathioprine

- For mild-moderate lupus when therapy with HCQ (+/- glucocorticoids) is ineffective.
- Also used as maintenance therapy after induction with more potent agents in patients with severe lupus affecting the kidneys, nervous system, or patients with vasculitis

Methotrexate

- For mild-moderate lupus when therapy with HCQ (+/- glucocorticoids) is ineffective
- Effective in refractory arthritis, skin disease, myositis, pleuritis, or pericarditis

Cyclophosphamide

- Inductive therapy to treat severe lupus involving the kidneys, nervous system, or vasculitis
- Used to minimize damage and induce remission

OTHER PHARMACOLOGICAL THERAPIES (Continued) Second-Line

Mycophenolate mofetil

- Equivalent to cyclophosphamide for induction in proliferative lupus nephritis
- Preferred to cyclophosphamide in patients who want to preserve fertility
- More effective than cyclophosphamide in patients with Hispanic or African backgrounds
- More effective (equally safe) compared to azathioprine for maintenance therapy in patients with nephritis

Rituximab

- Used for refractory disease (lupus nephritis or severe hematologic involvement)

OTHER PHARMACOLOGICAL THERAPIES (Continued) Second-Line

Belimumab and anifrolumab

- Belimumab: used in combination with standard therapies for treating active autoantibody-positive SLE or used in combination to treat and maintain refractory proliferative lupus nephritis
- Anifrolumab: used in combination with standard therapies for treatment of severe skin disease
- Considered for those who cannot tolerate or who are unresponsive to HCQ with or without glucocorticoids
  - Can be considered first-line in severe non-renal SLE, but with extensive disease

Tacrolimus and cyclosporine are calcineurin inhibitors

- Used to treat and maintain refractory proliferative lupus nephritis and severe lupus nephritis with proteinuria
- Cyclosporine is equivalent to azathioprine for remission maintenance following induction therapy
- Tacrolimus > cyclosporine due to AE profile (lower incidences of HTN, hyperlipidemia, gingival hyperplasia, hirsutism) (but limited long-term studies)

DOSAGE INFORMATION

Class	Drug	Side Effects	Comments
Antimalarials	Hydroxy-chloroquine sulfate	Nausea, cramps, diarrhea, rash, headache, skin deposition (hyperpigmentation). Rare retinal deposition and ocular toxicity (dose-related), myopathy. Rare reports of severe hypoglycemia with or without oral hypoglycemic agents	• May increase digoxin levels and may potentiate the effects of beta-blockers • Ophthalmologic assessments required Q1–5 years, depending on risk factors • Target dose is 5 mg/kg/day *(based on actual body weight and patient factors) and max. 400 mg/day*
B-Cell Depletors	Rituximab	Mild to severe infusion reactions (rare reports of very severe reactions leading to death) Rare: progressive multifocal leukoencephalopathy (PML)	• Premedicate with acetaminophen and antihistamine prior to infusion. Monitor for hypersensitivity reactions • Benefits of treatment may be delayed by 3–9 months

DOSAGE INFORMATION

Class	Drug	Side Effects	Comments
Corticosteroids oral	Prednisone	Acne, skin fragility, striae, GI upset, weight gain, glucose intolerance, mood swings, myopathy, glaucoma, cataracts, hypertension, osteoporosis, avascular necrosis, adrenal suppression, increased susceptibility to infections	• Drug interactions: increases risk of GI ulceration • Consider prophylaxis for drug-induced osteoporosis in patients taking initial dose of ≥30 mg/day or cumulative dose of >5 g/year
Corticosteroids injectable	Methyl-prednisolone sodium succinate		

DOSAGE INFORMATION

Class	Drug	Side Effects	Comments
B-Lymphocyte Stimulator-Specific Inhibitors	Belimumab (Benlysta®)	Nausea, diarrhea, fever, anxiety, insomnia, depression, infusion reactions, hypersensitivity	• Premedicate with acetaminophen and an antihistamine before infusion; monitor for hypersensitivity reactions • Safety and efficacy not evaluated in patients with severe active central nervous system lupus • Smoking may affect efficacy of belimumab
Monoclonal antibody	Anifrolumab (Saphnelo®)	Nasopharyngitis, UTI, upper respiratory infections, infusion reaction, headache, herpes zoster	• Premedicate with acetaminophen and an antihistamine before infusion • Safety and efficacy not evaluated in patients with severe active lupus nephritis or severe active central nervous system lupus

DOSAGE INFORMATION

Class	Drug	Side Effects	Comments
Immuno-modulators	Cyclophosphamide	Nausea, vomiting, cytopenias, infertility, hemorrhagic cystitis, increased susceptibility to infections, potential for malignancy	• Modified dosing is required in Caucasian populations
Immuno-modulators	Methotrexate	Nausea, malaise, headache, oral ulcers, alopecia, diarrhea, cytopenias, hepatotoxicity, pneumonitis	• Drug interactions: NSAIDs/penicillins: may increase MTX serum concentrations but not clinically significant; sulfonamides: may decrease MTX clearance; alcohol increases risk of hepatoxicity • Abortogenic and teratogenic • Increased risk of toxicity in patients with lupus nephritis and renal impairment

DOSAGE INFORMATION

Class	Drug	Side effects	Comments
Immuno-modulators	Mycophenolate mofetil	Anemia, leukopenia, thrombocytopenia, hyper/hypotension, edema, hyperglycemia, hypercholesteremia, hypokalemia, nausea, vomiting, diarrhea, abdominal pain, headache, dizziness, rash	• Drug interactions: antacids, iron, magnesium and cholestyramine decrease absorption; decreases efficacy of oral contraceptives; increased drug concentrations with probenecid • Monitor CBC, LFTs and creatinine • Mycophenolic acid is likely equivalent in efficacy to mycophenolate mofetil in SLE
Immuno-modulators	Azathioprine	Nausea, vomiting, diarrhea, fever, malaise, hepatotoxicity, increased LFTs, leukopenia, thrombocytopenia, infection, myalgias	• Drug interactions: allopurinol: may increase azathioprine toxicity, monitor & decrease dose to 25-33%; warfarin/other anticoagulants: hypothrombinemic response may be inhibited; ACEi: combination increases risk of neutropenia; use with immunosuppressants increases risk of infections • Takes up to 3 months to reach effectiveness • Monitor CBC, LFTs, and creatinine

SPECIAL POPULATIONS – PREGNANCY AND BREASTFEEDING

- In general, pregnancy should be avoided in active disease, especially if there is significant organ dysfunction because of an increased risk of miscarriage and flares
  - Ideally, SLE should be in remission for 6 months on pregnancy compatible medications before attempting conception
  - In women who are planning on becoming pregnant, several tests and screening should be conducted due to greater adverse pregnancy events

HCQ is the only medication that is recommended during pregnancy

Selective use
• AZA (max dose of 2 mg/kg/day) • Prednisone (<10 mg/day) • Cyclosporine • Tacrolimus

With caution
• Biologics

Contraindicated
• CYC during 1st trimester • Mycophenolate • Leflunomide • Methotrexate

- Breastfeeding is encouraged in most women; however, medication use should be personalized with specific risks reviewed
  - HCQ, AZA, prednisone, cyclosporine, tacrolimus and biologics are considered compatible
  - MTX, CYC, mycophenolate, leflunomide are not compatible

MONITORING PARAMETERS

- Symptom control/remission
- Organ function (e.g. renal function)
- Infection risk (e.g. screening, vaccinations)
- Cardiovascular risk (e.g. VTE, hypertension, MI)
- Drug specific side effects (e.g. ophthalmology assessment, osteoporosis, cancer screening)

TIPS/TAKEAWAY

- Review the immunization status prior to initiating immunosuppressive therapy. Live vaccines (e.g. MMR, varicella) are not recommended in immunosuppressed patients. Vaccinations should be kept up to date while on immunosuppressive therapy, particularly influenza and pneumococcal (inactivated vaccines)
- Assess for and manage CV risk factors in patients with SLE due to their increased risk of CVD
- Patients with SLE are at a higher risk for osteoporosis due to sun avoidance and long-term CS use. Consider calcium and vitamin D, as well as bisphosphonates to prevent CS-induced osteoporosis.
- Sulfamethoxazole/trimethoprim can induce disease flares and photosensitive rashes (use with caution)
- Estrogen-containing contraceptives should be avoided in those with antiphospholipid antibodies or a history of thrombosis
- Hormone replacement therapy is associated with risk of mild to moderate disease flares
- Follow-up is individualized based on disease stage, medication onset and comorbidities. Usually, at least every 6 months, once the disease has stabilized. No strong evidence for a certain timeframe.
- Local rashes can be treated with topical corticosteroids or calcineurin inhibitors
- Hydroxychloroquine is standard first-line for all patients with SLE, except when contraindicated
- Glucocorticoids can be used to induce remission and should be limited to short term
- Other pharmacological agents (second line) can be considered based on patient-specific factors

CASE STUDY

LR is a 28-year-old female who presents to the clinic with symptoms of persistent fatigue, intermittent joint pain in her hands and knees, occasional shortness of breath, and a photosensitive rash on her cheeks. She explains that these symptoms have occurred over the past six months, and she has also experienced two episodes of painless oral ulcers in the past month. Her past medical history is significant for migraines and allergic rhinitis. Her current medications include Mirena® (levonorgestrel 52 mg) 20 mcg/day as an intrauterine device, cetirizine 20 mg PO daily, olopatadine 0.1% 1 drop instilled into the affected eye(s) BID, and sumatriptan 50 mg PO as a single dose. LR's mother was diagnosed with rheumatoid arthritis at the age of 58. LR is diagnosed with systemic lupus erythematosus (SLE) with suspected lupus nephritis and is referred for a nephrology consult. LR's rheumatologist initiates hydroxychloroquine 200 mg PO daily, prednisone 30 mg PO daily, calcium 600 mg/vitamin D 500 units 2 tablets PO daily, and pantoprazole 40 mg PO daily. LR does not smoke, drink alcohol, or use recreational drugs. She has no known allergies to any medications. She weighs 136.4 lbs.

Her most recent laboratory results revealed positive results for antinuclear antibodies (ANA) and anti-double-stranded DNA (dsDNA), a hemoglobin level of 105 g/L (reference: 120 to 160 g/L), a serum creatinine level of 135 µmol/L (reference: 44 to 97 µmol/L), a complement C3 level of 0.55 g/L (reference: 1 to 2.3 g/L), a complement C4 level of 0.08 g/L (reference: 0.1 to 0.5 g/L), an erythrocyte sedimentation rate (ESR) of 75 mm/hr (reference: 0 to 20 mm/hr), and a C-reactive protein (CRP) level of 90.1 nmol/L (reference: ≤76.2 nmol/L). Urinalysis results reveal the presence of proteins and red blood cells.

CASE STUDY - QUESTIONS

1. Which of the following tests is the most important to perform before LR begins her prescribed therapy?
   a) Dual-energy X-ray absorptiometry scan
   b) Ophthalmologic assessment
   c) Fasting lipid panel
   d) Blood glucose control

2. What is the most accurate explanation for LR's nephrology referral?
   a) Elevated ESR and CRP
   b) Prednisone use
   c) Urinalysis results
   d) Positive results for ANA and anti-dsDNA

3. Which of the following agents is LEAST appropriate to recommend for the treatment and prevention of LR's photosensitive rash?
   a) Topical pimecrolimus
   b) Topical desonide
   c) Sun protection factor (SPF) 50 cream
   d) Oral azathioprine

4. Which of the following vaccinations is NOT recommended for LR to receive?
   a) FluMist® Quadrivalent (influenza) nasal spray vaccine
   b) Capvaxive® (pneumococcal 21-valent conjugate) vaccine
   c) Shingrix® (herpes zoster) vaccine
   d) GARDASIL 9® (human papillomavirus 9-valent) vaccine

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CHANGE LOG

May 2025

- Slides 12, 21: Removed ASA

August 2025

- Formatting, grammar, spelling changes made throughout
- Slide 3: Updated legend
- Slide 5: Added information to weight change causes
- Slide 9: Changed 'IU' to 'units'
- Slides 26, 27: Added case study and questions

November 2025

- Slide 3: Updated legend
- Slide 25: Changed slide title to 'Takeaway' and added information

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 4: Added the names of the 4 types of lupus
- Slide 12: Minor update
- Slides 14, 15, 17: Updated information
- Slide 25: Consolidated info, minor update on glucocorticoids
- Slide 28: Updated references
- Formatting changes made